The Chain Rule

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Basic chain rule applications

Differentiate each function using the chain rule:

a $f(x) = (4x - 1)^5$

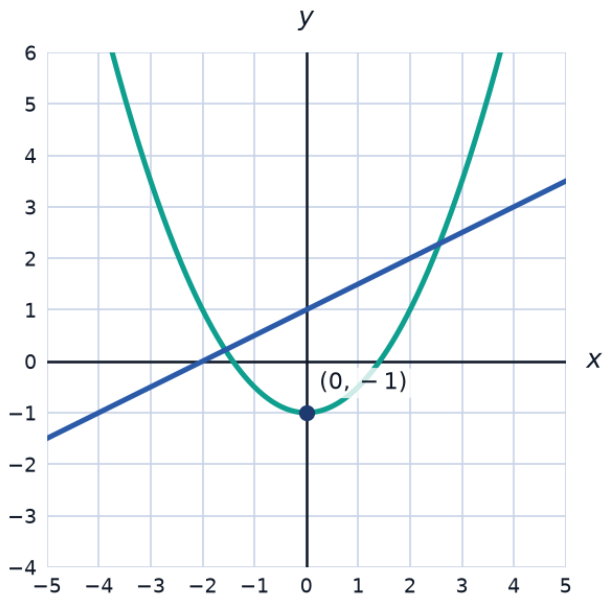
b $f(x) = \sin(3x^2)$

c $f(x) = e^{5x}$

d $f(x) = \ln(2x + 7)$

2 [Foundational] • Target: Basic chain rule applications

Differentiate $f(x) = \sqrt{x^2 + 9}$.



3 [Foundational] • Target: The chain rule with trig and exponential functions

Differentiate $f(x) = \cos^3(x)$ (i.e. $[\cos x]^3$).

4 [Foundational] • Target: The chain rule with trig and exponential functions

Differentiate $f(x) = e^{\sin x}$.

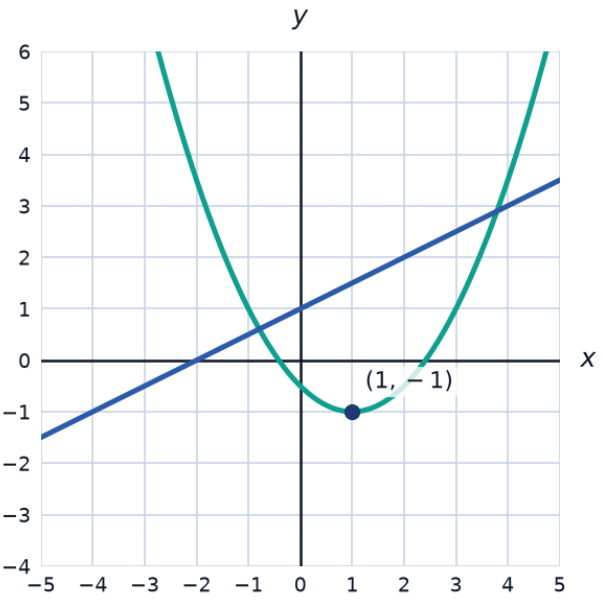
5 [Foundational] • Target: The chain rule with trig and exponential functions

Differentiate $f(x) = \tan(e^x)$.

© ClipSAT — clipsat.org | All Rights Reserved

6 [Foundational] • Target: Nested (multiple) chain rule

Differentiate $f(x) = \sin\left(\sqrt{x^2 + 1}\right)$, applying the chain rule twice.



7 [Foundational] • Target: Nested (multiple) chain rule

Differentiate $f(x) = (\ln(3x))^4$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Chain rule combined with product/quotient rules

Differentiate $f(x) = x^2 e^{3x}$.

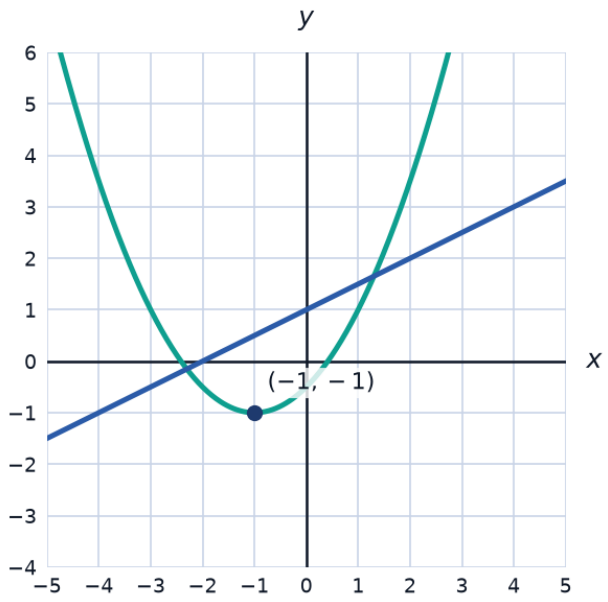
9 [Intermediate] • Target: Chain rule combined with product/quotient rules

Differentiate $f(x) = \frac{\sin(2x)}{x}$.

© ClipSAT — clipsat.org | All Rights Reserved

10 **[Intermediate]** • **Target: Chain rule with tables of values**

Given $f(2) = 5$, $f'(2) = 3$, $g(2) = 7$, $g'(2) = -1$, find $\frac{d}{dx}[f(g(x))]$ at $x = 2$ if $g(2) = 7$ and $f'(7) = 4$ (note the composition uses f' evaluated at $g(2) = 7$, not at 2).



11 **[Intermediate]** • **Target: Chain rule with inverse trig and logarithms**

Differentiate $f(x) = \sin^{-1}(2x)$.

12 **[Intermediate]** • **Target: Chain rule with inverse trig and logarithms**

Differentiate $f(x) = \ln(\sin x)$.

13 **[Intermediate]** • **Target: Chain rule with inverse trig and logarithms**

Differentiate $f(x) = \tan^{-1}(x^2)$.

14 **[Intermediate]** • **Target: Finding tangent lines using the chain rule**

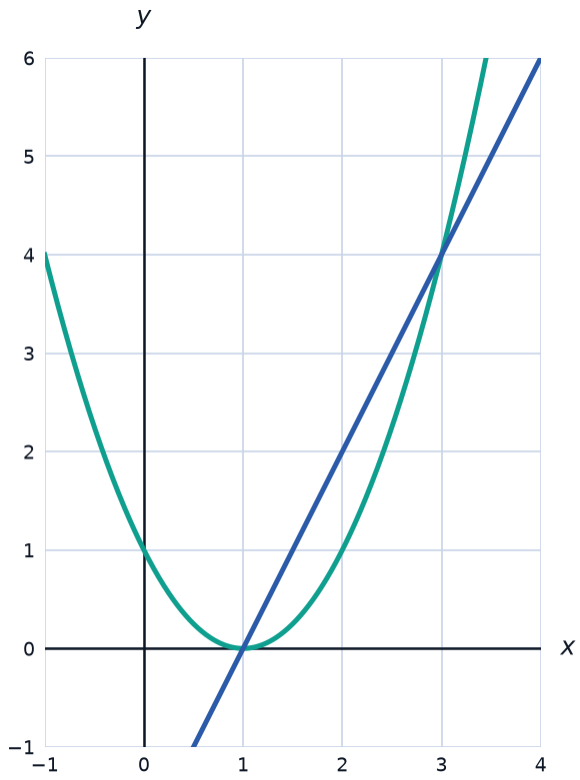
Find the equation of the tangent line to $f(x) = (2x - 1)^3$ at $x = 1$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

© ClipSAT — clipsat.org | All Rights Reserved

15 **[Challenging] • Target: Visualizing a composite function and its derivative • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions**

The graph shows $f(x) = (x - 1)^2$ (an inner-outer composite of $u = x - 1$ and u^2), with its tangent line at $x = 2$.



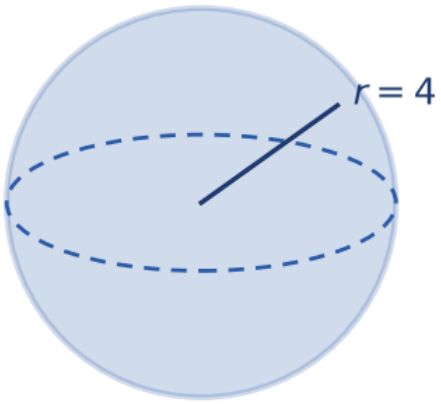
a Use the chain rule to find $f'(x)$.

b Confirm $f'(2)$ matches the slope of the tangent line shown.

© ClipSAT — clipsat.org | All Rights Reserved

16 [Challenging] • Target: The chain rule in applied rate problems • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

The volume of a sphere is $V = \frac{4}{3}\pi r^3$, and the radius is growing according to $r(t) = 2t + 1$ (cm, seconds). Use the chain rule to find $\frac{dV}{dt}$ at $t = 2$.



17 [Challenging] • Target: Chain rule with a table of composed values • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

Given $h(x) = f(g(x))$ where $g(1) = 3$, $g'(1) = 2$, $f(3) = 7$, $f'(3) = -4$, find $h'(1)$.

18 [Challenging] • Target: Chain rule with a table of composed values • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Differentiate $f(x) = \sqrt[3]{x^2 + 1}$ using the chain rule.

19 [Challenging] • Target: Chain rule with tables of values • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Given $g(6) = -5$, $g'(6) = -3$, and $f'(-5) = 6$, find $h'(6)$ if $h(x) = f(g(x))$.

20 [Challenging] • Target: Chain rule with inverse trig and logarithms • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Given $g(2) = -5$, $g'(2) = 1$, and $f'(-5) = -5$, find $h'(2)$ if $h(x) = f(g(x))$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • Target: Basic chain rule applications

Use the chain rule to differentiate $f(x) = (x - 1)^2$, then find $f'(1)$.

a. 0

b. -2

c. 3

d. 5

22 [Foundational] • Target: The chain rule with trig and exponential functions

Use the chain rule to differentiate $f(x) = (4x - 3)^2$, then find $f'(1)$.

a. 4

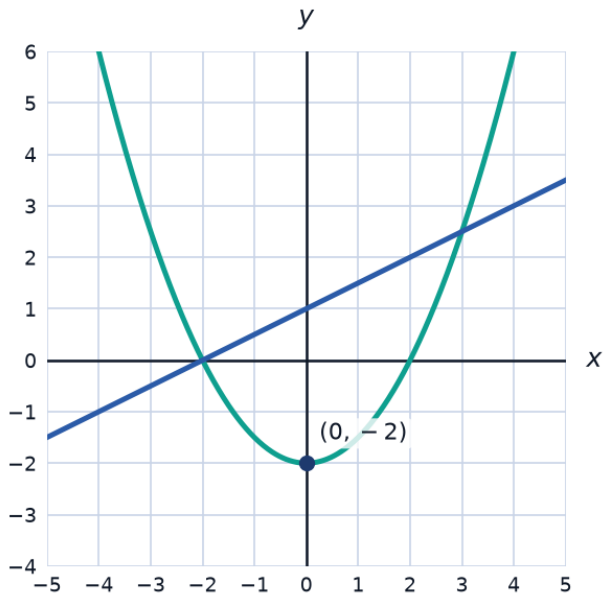
b. 6

c. 2

d. 8

23 [Foundational] • Target: Nested (multiple) chain rule

Use the chain rule to differentiate $f(x) = (4x - 1)^2$, then find $f'(1)$.



a. 24

b. 72

c. 12

d. 6

24 [Foundational] • Target: Chain rule combined with product/quotient rules

Use the chain rule to differentiate $f(x) = (3x)^4$, then find $f'(1)$.

a. 81

b. 324

c. 108

d. 972

© ClipSAT — clipsat.org | All Rights Reserved

25 [Foundational] • Target: Chain rule with tables of values

Use the chain rule to differentiate $f(x) = (2x + 3)^4$, then find $f'(0)$.

a. 108

b. 648

c. 216

d. 54

26 [Foundational] • Target: Chain rule with inverse trig and logarithms

Use the chain rule to differentiate $f(x) = (x - 1)^3$, then find $f'(0)$.

a. 3

b. -3

c. 0

d. 1

27 [Foundational] • Target: Finding tangent lines using the chain rule

Use the chain rule to differentiate $f(x) = (2x + 4)^2$, then find $f'(-2)$.

a. -2

b. 0

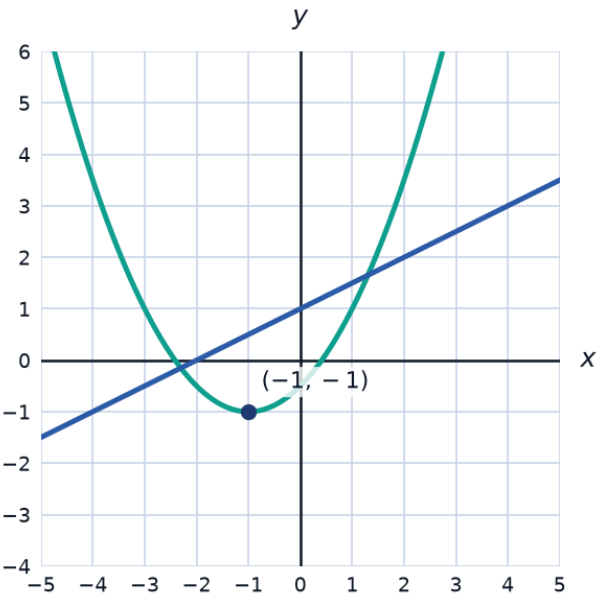
c. 2

d. 5

© ClipSAT — clipsat.org | All Rights Reserved

28 [Foundational] • Target: Visualizing a composite function and its derivative

Given $g(5) = 4$, $g'(5) = -2$, and $f'(4) = 5$, find $h'(5)$ if $h(x) = f(g(x))$.



a. 5

b. -2

c. 3

d. -10

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: The chain rule in applied rate problems

Given $g(6) = -5$, $g'(6) = -6$, and $f'(-5) = -4$, find $h'(6)$ if $h(x) = f(g(x))$.

a. -6

b. 24

c. -4

d. -10

30 [Intermediate] • Target: Chain rule with a table of composed values

Given $g(4) = -3$, $g'(4) = 6$, and $f'(-3) = 3$, find $h'(4)$ if $h(x) = f(g(x))$.

a. 9

b. 18

c. 6

d. 3

© ClipSAT — clipsat.org | All Rights Reserved

31 [Intermediate] • Target: Basic Chain Rule & Composite Functions

Given $g(7) = -4$, $g'(7) = 3$, and $f'(-4) = 5$, find $h'(7)$ if $h(x) = f(g(x))$.

a. 8

b. 5

c. 3

d. 15

32 [Intermediate] • Target: Exponential & Logarithmic Chain Rule

Given $g(3) = -5$, $g'(3) = 6$, and $f'(-5) = 2$, find $h'(3)$ if $h(x) = f(g(x))$.

a. 2

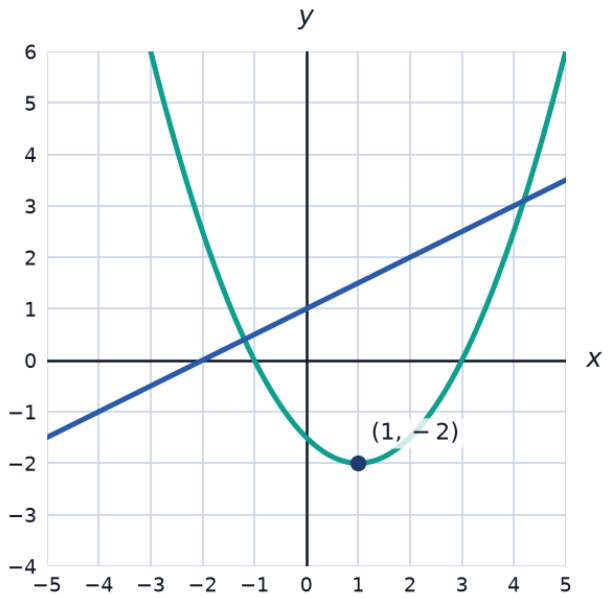
b. 6

c. 12

d. 8

33 [Intermediate] • Target: Repeated Chain Rule Applications

Given $g(3) = -6$, $g'(3) = -2$, and $f'(-6) = -3$, find $h'(3)$ if $h(x) = f(g(x))$.



a. -5

b. -3

c. 6

d. -2

34 [Intermediate] • Target: Table-Based Chain Rule Problems

Use the chain rule to differentiate $f(x) = \sqrt{x^2 + 16}$, then find $f'(-3)$.

a. $-\frac{5}{3}$

b. $-\frac{3}{4}$

c. $-\frac{3}{5}$

d. $\frac{3}{5}$

© ClipSAT — clipsat.org | All Rights Reserved

35 [Intermediate] • Target: Basic chain rule applications

Use the chain rule to differentiate $f(x) = \sqrt{x^2 + 144}$, then find $f'(-5)$.

a. $-\frac{5}{12}$

b. $-\frac{13}{5}$

c. $-\frac{5}{13}$

d. $\frac{5}{13}$

36 [Intermediate] • Target: The chain rule with trig and exponential functions

Given $h(2) = -3$, $h'(2) = -3$, $g(-3) = -1$, $g'(-3) = -4$, and $f'(-1) = -3$, find $k'(2)$ if $k(x) = f(g(h(x)))$.

a. -36

b. -10

c. 12

d. 9

37 [Intermediate] • Target: Nested (multiple) chain rule

Given $h(5) = 0$, $h'(5) = -5$, $g(0) = -5$, $g'(0) = 3$, and $f'(-5) = -4$, find $k'(5)$ if $k(x) = f(g(h(x)))$.

a. 60

b. -12

c. -6

d. 20

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Chain rule combined with product/quotient rules • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Given $h(2) = 0$, $h'(2) = 2$, $g(0) = 1$, $g'(0) = -4$, and $f'(1) = -5$, find $k'(2)$ if $k(x) = f(g(h(x)))$.

a. 40

b. -10

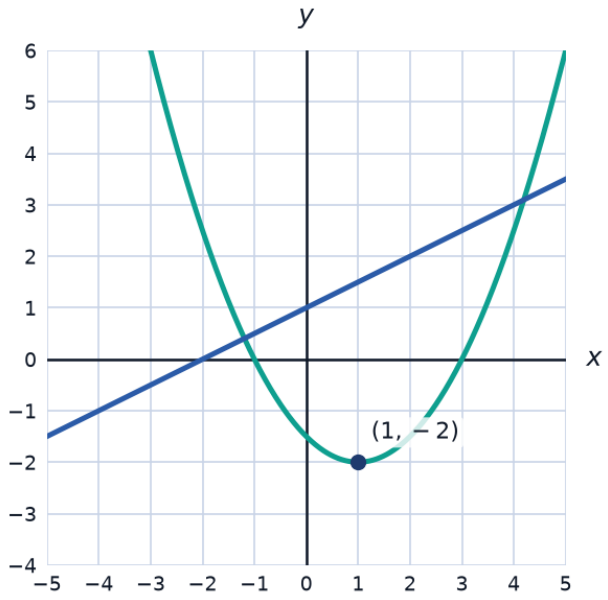
c. 20

d. -7

© ClipSAT — clipsat.org | All Rights Reserved

39 [Challenging] • Target: Chain rule with tables of values • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Which is the correct derivative of $F(x) = \sin(-2x - 1)$?



a. $-2\cos(-2x - 1)$

b. $2\sin(-2x - 1)$

c. $-2\cos(x)$

d. $\cos(-2x - 1)$

40 [Challenging] • Target: Chain rule with inverse trig and logarithms • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Which is the correct derivative of $F(x) = \sin(-6x + 4)$?

a. $-6\cos(x)$

b. $-6\cos(-6x + 4)$

c. $\cos(-6x + 4)$

d. $6\sin(-6x + 4)$

41 [Challenging] • Target: Basic Chain Rule & Composite Functions • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

If $h(x) = f(g(x))$ with $g(3) = 5, g'(3) = -2,$ and $f'(5) = 7$, what is $h'(3)$?

a. -14

b. 10

c. -10

d. 35

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Exponential & Logarithmic Chain Rule • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Find the derivative of $f(x) = e^{5x^2 - 2x}$.

a. $(10x - 2)e^{5x^2 - 2x}$

b. $e^{10x - 2}$

c. $10xe^{5x^2 - 2x}$

d. $(5x^2 - 2x)e^{5x^2 - 2x}$

43 [Challenging] • Target: Exponential & Logarithmic Chain Rule • Cross-Lesson: 8.01 Area Between Curves

Find $\frac{d}{dx}[\ln(\cos x)]$.

a. $-\tan x$

b. $\tan x$

c. $\cot x$

d. $-\cot x$

44 [Challenging] • Target: Repeated Chain Rule Applications • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Differentiate $f(x) = \sqrt{1 + \tan(3x)}$.

a. $\frac{3\sec^2(3x)}{2\sqrt{1 + \tan(3x)}}$

b. $\frac{\sec^2(3x)}{2\sqrt{1 + \tan(3x)}}$

c. $\frac{3\sec^2(3x)}{\sqrt{1 + \tan(3x)}}$

d. $\frac{1}{2\sqrt{1 + \tan(3x)}}$

45 [Challenging] • Target: Table-Based Chain Rule Problems • Cross-Lesson: 1.01 Limits Graphically and Numerically

Let $h(x) = [f(x)]^3$. If $f(1) = 2$ and $f'(1) = -4$, what is $h'(1)$?

a. -48

b. -24

c. 12

d. -16

© ClipSAT — clipsat.org | All Rights Reserved